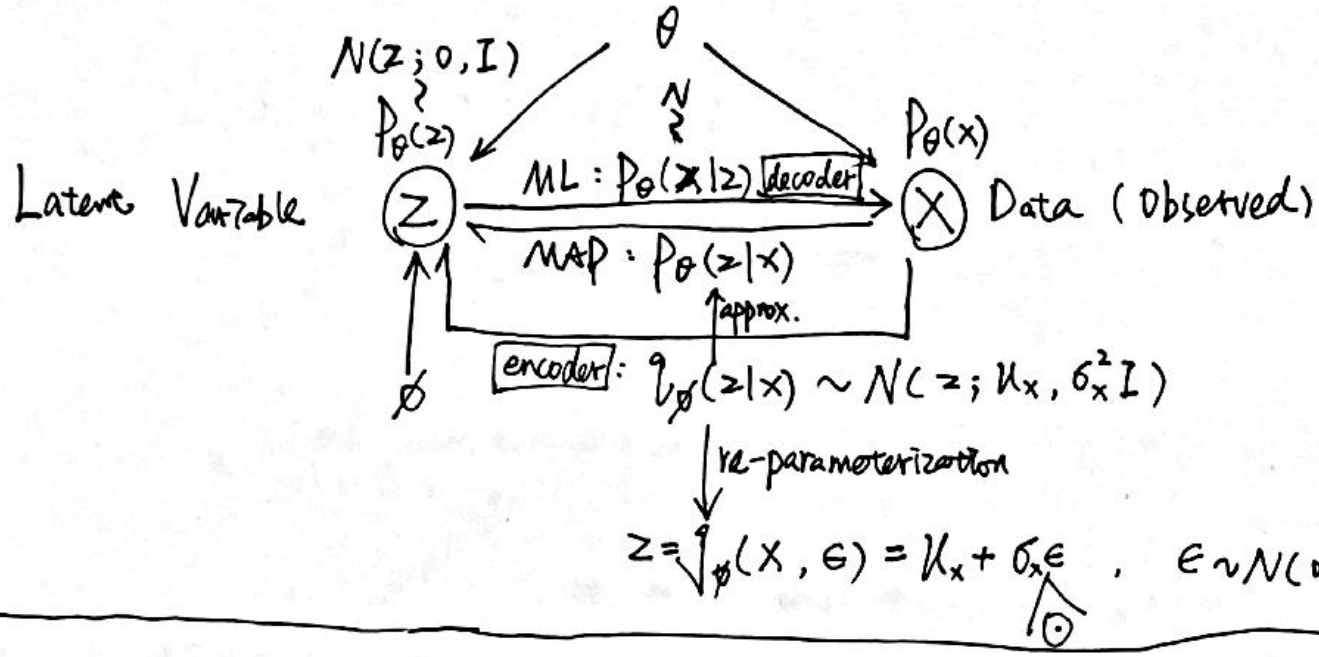
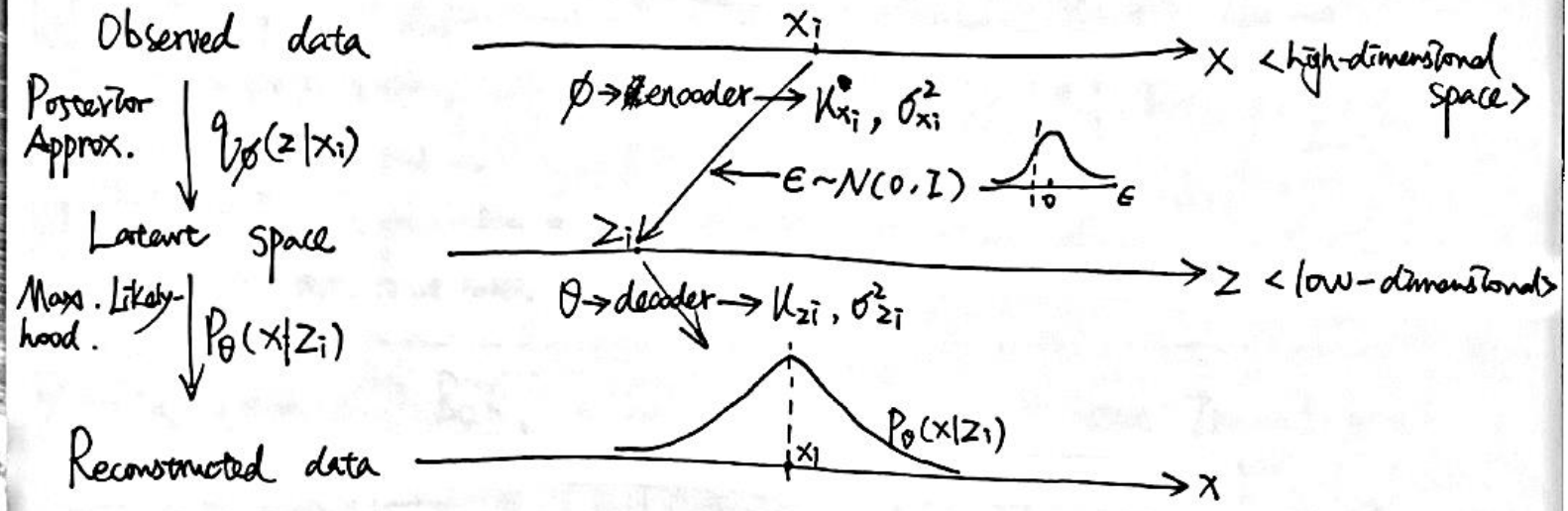
Variational Auto-Encoder





Training target: $\log P(x_i) = \int_z q_\phi(z|x_i) \log P_\theta(x_i) dz$

<for single sample>

$$= \int_z q_\phi(z|x_i) \log \frac{P_\theta(x_i|z) P_\theta(z)}{P_\theta(z|x_i)} dz$$

$$= \int_z q_\phi(z|x_i) \log \frac{P_\theta(x_i|z) P_\theta(z)}{q_\phi(z|x_i)} dz + \int_z q_\phi(z|x_i) \log \frac{q_\phi(z|x_i)}{P_\theta(z|x_i)} dz$$

$$= VLB + D_{KL}(q_\phi(z|x_i) \,\|\, P_\theta(z|x_i)) \geq VLB$$

Real target: $VLB = -\int_z q_\phi(z|x_i) \log \frac{q_\phi(z|x_i)}{P_\theta(z)} dz + \int_z q_\phi(z|x_i) \log P_\theta(x_i|z) dz$

$$= -D_{KL}(q_\phi(z|x_i) \,\|\, P_\theta(z)) + E_{q_\phi(z|x_i)}[\log P_\theta(x_i|z)]$$

$$\approx \frac{1}{L} \sum_{l=1}^{L} (\log P_\theta(x_i|z_{i,l})),$$

$$z_{i,l} = g_\phi(\epsilon_{i,l}, x_i), \quad \epsilon_{i,l} \sim P(\epsilon)$$

encourage $q_\phi(z|x_i)$ to be close to the prior $P_\theta(z)$

Mini-Batch, "ϕ, θ" optimization ...